In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
dataset=pd.read_csv("mall_customers.csv")
dataset
```

Out[2]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [4]:
```
x=dataset.iloc[:,3:5].values
print(x)
```

```
[[ 15  39]
 [ 15  81]
 [ 16   6]
 [ 16  77]
 [ 17  40]
 [ 17  76]
 [ 18   6]
 [ 18  94]
 [ 19   3]
 [ 19  72]
 [ 19  14]
 [ 19  99]
 [ 20  15]
 [ 20  77]
 [ 20  13]
 [ 20  79]
 [ 21  35]
 [ 21  66]
 [ 23  29]
 [ 23  98]
 [ 24  35]
 [ 24  73]
 [ 25   5]
 [ 25  73]
 [ 28  14]
 [ 28  82]
 [ 28  32]
 [ 28  61]
 [ 29  31]
 [ 29  87]
 [ 30   4]
 [ 30  73]
 [ 33   4]
 [ 33  92]
 [ 33  14]
 [ 33  81]
 [ 34  17]
 [ 34  73]
 [ 37  26]
 [ 37  75]
 [ 38  35]
 [ 38  92]
 [ 39  36]
 [ 39  61]
 [ 39  28]
 [ 39  65]
 [ 40  55]
 [ 40  47]
 [ 40  42]
 [ 40  42]
 [ 42  52]
 [ 42  60]
 [ 43  54]
 [ 43  60]
 [ 43  45]
 [ 43  41]
 [ 44  50]
 [ 44  46]
 [ 46  51]
 [ 46  46]
 [ 46  56]
```

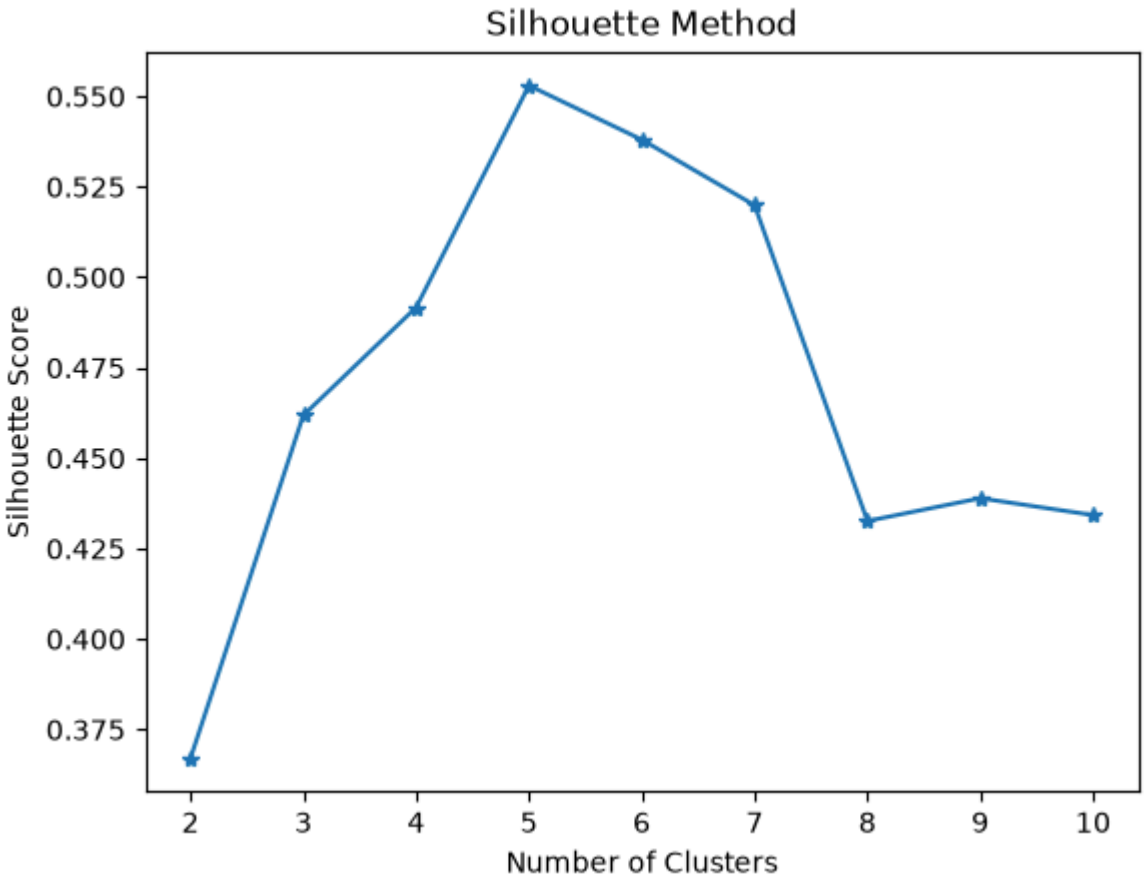
```
[ 46  55]
[ 47  52]
[ 47  59]
[ 48  51]
[ 48  59]
[ 48  50]
[ 48  48]
[ 48  59]
[ 48  47]
[ 49  55]
[ 49  42]
[ 50  49]
[ 50  56]
[ 54  47]
[ 54  54]
[ 54  53]
[ 54  48]
[ 54  52]
[ 54  42]
[ 54  51]
[ 54  55]
[ 54  41]
[ 54  44]
[ 54  57]
[ 54  46]
[ 57  58]
[ 57  55]
[ 58  60]
[ 58  46]
[ 59  55]
[ 59  41]
[ 60  49]
[ 60  40]
[ 60  42]
[ 60  52]
[ 60  47]
[ 60  50]
[ 61  42]
[ 61  49]
[ 62  41]
[ 62  48]
[ 62  59]
[ 62  55]
[ 62  56]
[ 62  42]
[ 63  50]
[ 63  46]
[ 63  43]
[ 63  48]
[ 63  52]
[ 63  54]
[ 64  42]
[ 64  46]
[ 65  48]
[ 65  50]
[ 65  43]
[ 65  59]
[ 67  43]
[ 67  57]
[ 67  56]
[ 67  40]
```

```
[ 69  58]
[ 69  91]
[ 70  29]
[ 70  77]
[ 71  35]
[ 71  95]
[ 71  11]
[ 71  75]
[ 71   9]
[ 71  75]
[ 72  34]
[ 72  71]
[ 73   5]
[ 73  88]
[ 73   7]
[ 73  73]
[ 74  10]
[ 74  72]
[ 75   5]
[ 75  93]
[ 76  40]
[ 76  87]
[ 77  12]
[ 77  97]
[ 77  36]
[ 77  74]
[ 78  22]
[ 78  90]
[ 78  17]
[ 78  88]
[ 78  20]
[ 78  76]
[ 78  16]
[ 78  89]
[ 78   1]
[ 78  78]
[ 78   1]
[ 78  73]
[ 79  35]
[ 79  83]
[ 81   5]
[ 81  93]
[ 85  26]
[ 85  75]
[ 86  20]
[ 86  95]
[ 87  27]
[ 87  63]
[ 87  13]
[ 87  75]
[ 87  10]
[ 87  92]
[ 88  13]
[ 88  86]
[ 88  15]
[ 88  69]
[ 93  14]
[ 93  90]
[ 97  32]
[ 97  86]
[ 98  15]
```

```
[ 98  88]
[ 99  39]
[ 99  97]
[101  24]
[101  68]
[103  17]
[103  85]
[103  23]
[103  69]
[113   8]
[113  91]
[120  16]
[120  79]
[126  28]
[126  74]
[137  18]
[137  83]]
```

In [15]:

```
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import silhouette_score
list=[]
for i in range(2,11):
    ac=AgglomerativeClustering(n_clusters=i,linkage="ward").fit(x)
    labels = ac.fit_predict(x)
    score = silhouette_score(x, labels)
    list.append(score)
plt.plot(range(2, 11), list, marker="*")
plt.xlabel("Number of Clusters")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Method")
plt.show()
```



In [17]:

```
ac=AgglomerativeClustering(n_clusters=5,linkage="ward")
```

```
y=ac.fit_predict(x)
print(y)
```

```
[4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4 3 4
 3 4 3 4 3 4 1 4 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 2 1 2 1 2 0 2 0 2 1 2 0 2 0 2 0 2 0 2 1 2 0 2 1 2
 0 2 0 2 0 2 0 2 0 2 0 2 1 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2
 2 0 2 0 2 0 2 0 2 0 2 0 2 0 2]
```

In [18]:
```
supervised=dataset
supervised
```

Out[18]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [19]:
```
supervised['cluster_group']=y
supervised
```

Out[19]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)	cluster_group
0	1	Male	19	15	39	4
1	2	Male	21	15	81	3
2	3	Female	20	16	6	4
3	4	Female	23	16	77	3
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	2
196	197	Female	45	126	28	0
197	198	Male	32	126	74	2
198	199	Male	32	137	18	0
199	200	Male	30	137	83	2

200 rows × 6 columns

In [20]:
```
supervised.to_csv('cluster1.csv',index=True)
dir(y)
```

Out[20]:

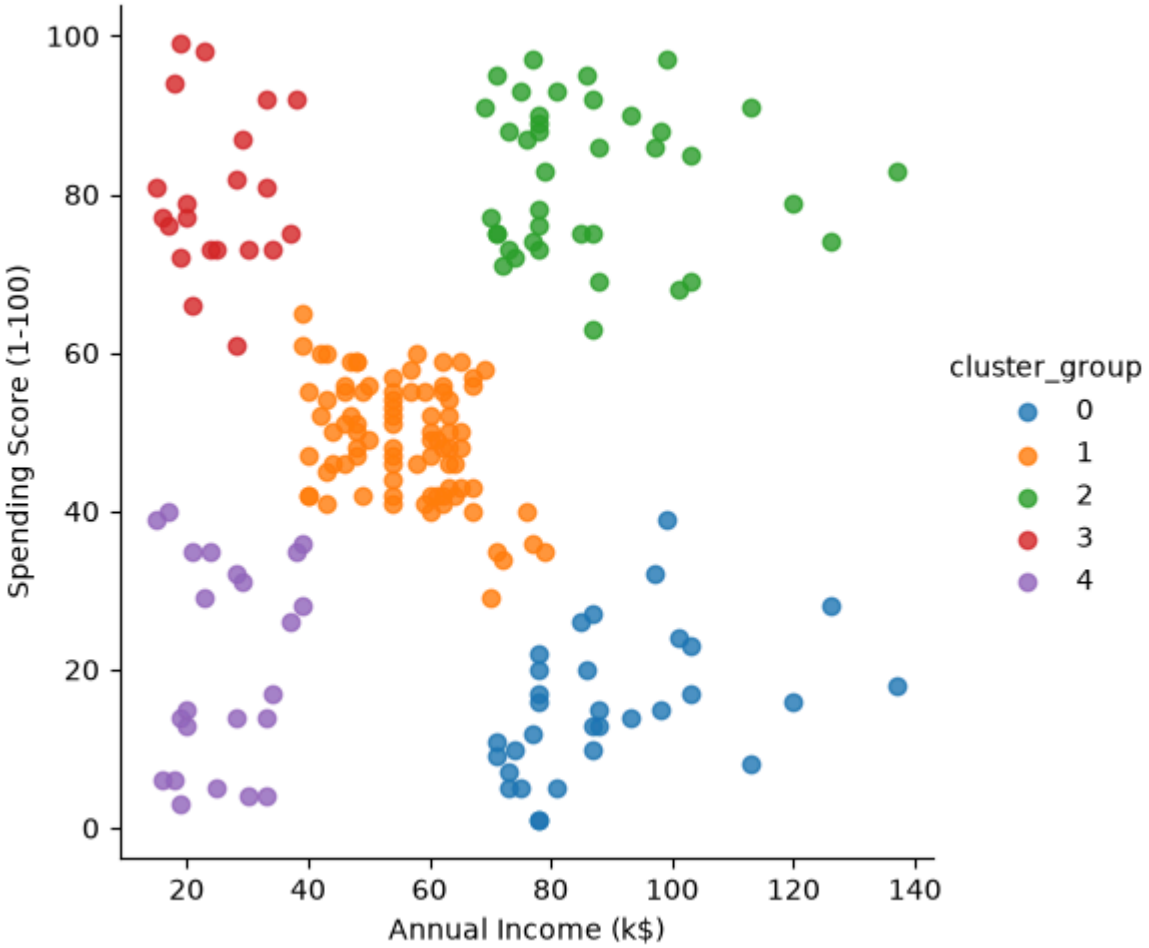
```
['T',
 '__abs__',
 '__add__',
 '__and__',
 '__array__',
 '__array_finalize__',
 '__array_function__',
 '__array_interface__',
 '__array_namespace__',
 '__array_priority__',
 '__array_struct__',
 '__array_ufunc__',
 '__array_wrap__',
 '__bool__',
 '__buffer__',
 '__class__',
 '__class_getitem__',
 '__complex__',
 '__contains__',
 '__copy__',
 '__deepcopy__',
 '__delattr__',
 '__delitem__',
 '__dir__',
 '__divmod__',
 '__dlpack__',
 '__dlpack_device__',
 '__doc__',
 '__eq__',
 '__float__',
 '__floordiv__',
 '__format__',
 '__ge__',
 '__getattribute__',
 '__getitem__',
 '__getstate__',
 '__gt__',
 '__hash__',
 '__iadd__',
 '__iand__',
 '__ifloordiv__',
 '__ilshift__',
 '__imatmul__',
 '__imod__',
 '__imul__',
 '__index__',
 '__init__',
 '__init_subclass__',
 '__int__',
 '__invert__',
 '__ior__',
 '__ipow__',
 '__irshift__',
 '__isub__',
 '__iter__',
 '__itruediv__',
 '__ixor__',
 '__le__',
 '__len__',
 '__lshift__',
 '__lt__',
```

```
'__matmul__',
'__mod__',
'__mul__',
'__ne__',
'__neg__',
'__new__',
'__or__',
'__pos__',
'__pow__',
'__radd__',
'__rand__',
'__rdivmod__',
'__reduce__',
'__reduce_ex__',
'__repr__',
'__rfloordiv__',
'__rlshift__',
'__rmatmul__',
'__rmod__',
'__rmul__',
'__ror__',
'__rpow__',
'__rrshift__',
'__rshift__',
'__rsub__',
'__rtruediv__',
'__rxor__',
'__setattr__',
'__setitem__',
'__setstate__',
'__sizeof__',
'__str__',
'__sub__',
'__subclasshook__',
'__truediv__',
'__xor__',
'all',
'any',
'argmax',
'argmin',
'argpartition',
'argsort',
'astype',
'base',
'byteswap',
'choose',
'clip',
'compress',
'conj',
'conjugate',
'copy',
'ctypes',
'cumprod',
'cumsum',
'data',
'device',
'diagonal',
'dot',
'dtype',
'dump',
'dumps',
```

```
'fill',
'flags',
'flat',
'flatten',
'getfield',
'imag',
'item',
'itemsize',
'mT',
'max',
'mean',
'min',
'nbytes',
'ndim',
'nonzero',
'partition',
'prod',
'put',
'ravel',
'real',
'repeat',
'reshape',
'resize',
'round',
'searchsorted',
'setfield',
'setflags',
'shape',
'size',
'sort',
'squeeze',
'std',
'strides',
'sum',
'swapaxes',
'take',
'to_device',
'tobytes',
'tofile',
'tolist',
'trace',
'transpose',
'var',
'view']
```

In [22]:
```
import seaborn as sns
facet = sns.lmplot(data=supervised, x=supervised.columns[3],
                   y=supervised.columns[4], hue=supervised.columns[5],
                   fit_reg=False,legend_out=True)
```

```
C:\Users\User\anaconda3\Lib\site-packages\seaborn\regression.py:598: UserWarning: legend_out is
deprecated from the `lmplot` function signature. Please update your code to pass it using
`facet_kws`.
  warnings.warn(msg, UserWarning)
```



In [ ]: